

Chi-Square (χ^2) Test

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Introduction

- Parametric tests (t , F , Z) assume normal distribution.
- These tests depend on population parameters.
- Such tests are called **parametric tests**.
- When assumptions are not valid, **Non-parametric tests** are used.

Non-Parametric Tests

- Distribution-free tests.
- Widely used after 1945.
- Popular in behavioral sciences.

Reasons for popularity:

- 1 No assumption about distribution.
- 2 Computationally simple.
- 3 Suitable for non-metric data.

χ^2 Test Definition

- Introduced by Karl Pearson (1900).
- Measures discrepancy between observed and expected frequencies.

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

- O = Observed frequency
- E = Expected frequency

Steps to Calculate χ^2

Step 1: Expected Frequency

$$E = \frac{RT \times CT}{N}$$

- RT = Row Total
- CT = Column Total
- N = Total observations

Step 2: Compute $(O - E)^2$

Step 3: Compute $\sum \frac{(O-E)^2}{E}$

Degrees of Freedom

$$\nu = n - k$$

- n = Number of classes
- k = Number of constraints

Contingency Table:

$$\nu = (r - 1)(c - 1)$$

Important Points

- $\sum(O - E) = 0$
- $\chi^2 \geq 0$
- Range: 0 to ∞
- Unimodal distribution

Large Degrees of Freedom ($\nu > 30$)

$$Z = \frac{\chi^2 - \nu}{\sqrt{2\nu}}$$

- Approximates standard normal distribution
- Standard deviation = 1

Conditions for χ^2 Test

- Sample size N should be sufficiently large.
- Expected frequency in each cell ≥ 5 .
- Constraints must be linear.

- χ^2 is a non-parametric test.
- Used for:
 - Test of independence
 - Goodness of fit
- Simple and widely applicable.

χ^2 Test as Test of Independence

- Used to determine whether two attributes are associated.
- Null Hypothesis (H_0): Attributes are independent.
- If calculated χ^2

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table value $\rightarrow$ Accept H_0

- If calculated χ^2

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table value $\rightarrow$ Reject H_0

Note:

- χ^2 does NOT measure strength of relationship.
- It only tests whether association exists.

χ^2 Test as Goodness of Fit

- Tests whether observed distribution fits theoretical distribution.
- Commonly used for:
 - Normal distribution
 - Binomial distribution
 - Poisson distribution

Null Hypothesis (H_0): Observed frequencies follow the specified theoretical distribution.

Steps in Goodness of Fit Test

- 1 State H_0 and H_1 .
- 2 Select level of significance (usually 5%).
- 3 Compute expected frequencies.
- 4 Calculate $\chi^2 = \sum \frac{(O-E)^2}{E}$.
- 5 Compare with table value.

Decision Rule:

- If $\chi^2_{calc} < \chi^2_{table} \rightarrow$ Good fit
- If $\chi^2_{calc} > \chi^2_{table} \rightarrow$ Poor fit

Degrees of Freedom (Goodness of Fit)

$$\nu = (k - 1 - m)$$

- k = Number of classes
- m = Number of estimated parameters

Example (Normal Distribution):

- If $k = 7$
- Estimate mean and standard deviation ($m = 2$)

$$\nu = 7 - 1 - 2 = 4$$

χ^2 Test of Homogeneity

- Tests whether different samples come from same population.
- Involves two or more independent samples.

Null Hypothesis (H_0): All samples are drawn from the same population.

Note: Same test statistic as independence test.

Independence vs Homogeneity

Feature	Independence	Homogeneity
Samples	One	Two or more
Purpose	Association	Compare populations
Data Type	Cross-classified	Cross-classified

Illustration 1: Quinine and Malaria

Given Data (2×2 Table)

- $N = 6496$
- Calculated $\chi^2 = 76.719$
- Degrees of freedom $\nu = 1$
- Table value at 5% = 3.84

Decision:

Since $76.719 > 3.84$, reject H_0 .

Conclusion: Quinine is effective in checking malaria.

Illustration 2: Ownership and Fertilizer Use

Given Data (2×2 Table)

- $N = 1000$
- Calculated $\chi^2 = 273.504$
- Degrees of freedom $\nu = 1$
- Table value at 5% = 3.84

Decision:

Since $273.504 > 3.84$, reject H_0 .

Conclusion: Owner cultivators are more inclined to use fertilizers.

Important Observation

- Extremely small χ^2 (too close to zero) may indicate manipulation.
- Both very large and very small values require careful interpretation.
- Always verify experimental design.

Chi-Square Test for Specified Population Variance

Purpose: To test whether a random sample comes from a normal population with a specified variance σ^2 .

Test Statistic:

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2}$$

where:

- S^2 = sample variance
- n = sample size
- σ^2 = population variance under H_0

Degrees of Freedom:

$$\nu = n - 1$$

Note: This test is valid only if the population is normally distributed.

Given:

- Sample size: $n = 25$
- Sample standard deviation: $S = 8.5$

Hypotheses:

$$H_0 : \sigma = 10$$

$$H_1 : \sigma \neq 10$$

Step 1: Compute the test statistic

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2} = \frac{24 \times (8.5)^2}{10^2} = \frac{24 \times 72.25}{100} = \frac{1734}{100} = 17.34$$

Step 2: Degrees of freedom

$$\nu = n - 1 = 24$$

Critical Value:

$$\chi^2_{0.05, 24} = 36.415$$

Decision:

$$\chi^2_{\text{calculated}} = 17.34 < 36.415$$

Conclusion: The null hypothesis is accepted. The population standard deviation may be 10.

Chi-Square Test: Examination Results

Given Data (O):

- Failed = 220
- Third Class = 170
- Second Class = 90
- First Class = 20

Total students = 500

General Examination Ratio:

4 : 3 : 2 : 1

Null Hypothesis (H_0): Observed results are commensurate with the given ratio.

Expected Frequencies (E):

$$\text{Failed} = \frac{500 \times 4}{10} = 200$$

$$\text{Third Class} = \frac{500 \times 3}{10} = 150$$

Chi-Square Calculation and Conclusion

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Category	O	E	$\frac{(O-E)^2}{E}$
Failed	220	200	2.000
Third Class	170	150	2.667
Second Class	90	100	1.000
First Class	20	50	18.000

$$\chi^2 = 23.667$$

Degrees of freedom:

$$\nu = 4 - 1 = 3$$

Critical value at 5% level:

$$\chi_{0.05,3}^2 = 7.81$$

Conclusion: Since $23.667 > 7.81$, reject H_0 . The results are **not in** accordance with the given ratio.

Movie Preference and Age Group

A movie producer wants to determine whether a new movie appeals equally to all age groups or more to a particular age group.

A random sample of people attending a preview show gave the following results:

Opinion	Under 20	20–39	40–59	60 & Over	Total
Liked the movie	146	78	48	28	300
Disliked the movie	54	22	42	22	140
Indifferent	20	10	10	20	60
Total	220	110	100	70	500

Question: What inference can be drawn from this data?

Chi-Square Test: Demand and Day of the Week

Problem Statement

The demand for a particular spare part in a factory was found to vary from day to day. To examine whether the demand depends on the day of the week, a sample study was conducted.

The following data were obtained:

Day	Mon.	Tues.	Wed.	Thurs.	Fri.	Sat.
No.of parts demanded	1124	1125	1110	1120	1126	1115

Hypothesis to be Tested: The number of parts demanded does not depend on the day of the week.

Given: Table value of χ^2 for 5 d.f. at 5% level of significance = 11.07

The F-Test (Variance Ratio Test)

The **F-test** is named in honour of the statistician **R. A. Fisher**.

Objective of the F-Test:

- To test whether two independent estimates of population variance differ significantly.
- To determine whether two samples are drawn from normal populations having the same variance.

Since the test is based on the ratio of two variances, it is also known as the **Variance Ratio Test**.

F-Test Statistic

The F-statistic is defined as:

$$F = \frac{S_1^2}{S_2^2}$$

where:

- S_1^2 = larger sample variance
- S_2^2 = smaller sample variance

Degrees of Freedom:

$$v_1 = n_1 - 1 \quad (\text{for larger variance})$$

$$v_2 = n_2 - 1 \quad (\text{for smaller variance})$$

Decision Rule for F-Test

The calculated value of F is compared with the table value $F_{\alpha}(v_1, v_2)$ at a given level of significance.

Decision Rule:

- If $F_{\text{calculated}} > F_{\text{table}}$, reject H_0
- If $F_{\text{calculated}} \leq F_{\text{table}}$, accept H_0

Conclusion:

- Rejection implies unequal population variances
- Acceptance implies equal population variances

Assumptions of the F-Test

The F-test is based on the following assumptions:

- ① **Normality:** Observations in each sample are normally distributed.
- ② **Homogeneity of Variance:** Population variances are equal:

$$\sigma_1^2 = \sigma_2^2 = \dots$$

- ③ **Independence:** Errors (deviations from group means) are independent.

Applications of the F-Test

The F-test is commonly used to:

- Compare variances of two normal populations

The following illustrations demonstrate the application of the F-test.

Illustration 1: F-Test for Equality of Variances

Two random samples are drawn from two normal populations with the following values:

Sample A:

66, 67, 75, 76, 82, 84, 88, 90, 92

Sample B:

64, 66, 74, 78, 82, 85, 87, 92, 93, 95, 97

Test whether the two populations have the same variance at the 5% level of significance.

Given: $F_{0.05}$ for $v_1 = 10$ and $v_2 = 8$ is 3.36.

Illustration 2: F-Test

In a sample of 8 observations, the sum of squared deviations from the mean is 84.4. In another sample of 10 observations, the corresponding value is 102.6.

Test whether the difference between the variances is significant at the 5% level of significance.

Given:

- $F_{0.05}(7, 9) = 3.29$
- $F_{0.05}(8, 10) = 3.07$